

Practice Final

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Problem 1 Consider generating a random variable X such that

$$\mathbb{P}(X = i) = \frac{e^{-\lambda} \lambda^i / i!}{\sum_{j=0}^k e^{-\lambda} \lambda^j / j!}, \quad i = 0, 1, 2, \dots, k.$$

- (a) Derive the inverse transform method to generate X .
- (b) Suppose you run the following algorithm to generate X : First sample from a Poisson distribution with rate λ , and if this value falls into $\{0, 1, \dots, k\}$, you output it; otherwise you return to the sampling step and repeat until you obtain an output. Is this an acceptance-rejection procedure? If it is, state precisely all the algorithmic configurations in this procedure.

Problem 2 Let $\alpha, \beta > 0$. Give a method for generating a random variable having distribution function

$$F(x) = 1 - \exp(-\alpha x^\beta), 0 < x < \infty.$$

- (a) Derive an inverse transform method to generate X .
- (b) Suppose you insist to use an acceptance-rejection method to generate X , using a proposal that has the same distribution function as above but with a new $\hat{\beta}$ such that $\hat{\beta} > \beta$, and that your computer has an established routine to generate from this proposal distribution directly. Does this acceptance-rejection method work? Explain.

Problem 3. A highway department has enough salt to handle a total of 80 inches of snowfall. Suppose the daily amount of snow has a mean of 1.5 inches and a standard deviation of 0.3 inches.

- (a) Approximate the probability that the salt on hand will suffice for the next 50 days.
- (b) What theorem did you use in solving part (a)? Do you think the theorem is justified? Explain briefly.
- (c) Suppose the highway department now has the capability to adjust its amount of salt. How many inches of snowfall should it prepare its salt for so that it is enough for the next 50 days with 99.5% chance?

Problem 4. Suppose that $X_1, \dots, X_n$ are normal with mean μ_1 ; $Y_1, \dots, Y_n$ are normal with mean μ_2 ; and $W_1, \dots, W_n$ are normal with mean $\mu_1 + \mu_2$. Assuming that all $3n$ random variables are independent with a common variance, find the maximum likelihood estimators of μ_1 and μ_2 .

Problem 5. A sample of 20 cigarettes is tested to determine nicotine content and the average value observed was 1.2 mg. You may assume that the nicotine content of each cigarette is normally distributed, and that the standard deviation of a cigarette's nicotine content is $\sigma = 0.2$ mg.

- (a) Compute a 95% confidence interval for the mean nicotine content of a cigarette.
- (b) Find the number of observations needed, in addition to the 20 already obtained, to ensure the point estimate is within ± 0.01 mg from the ground-truth mean nicotine content with 95% confidence.
- (c) Are your answers in parts (a) and (b) well justified by the assumptions? Explain briefly.